Streamlit application for FairFace

FairVision: Bias Detection and Mitigation in CNN-Based Age Classification

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Contents

1. Problem Definition and Dataset Overview

Face recognition systems are widely used in modern AI applications such as surveillance and authentication. However, these systems often suffer from **algorithmic bias**, where performance varies across demographic groups.

This project develops a CNN-based age classification system using the FairFace dataset. The goal is not only classification accuracy but also fairness evaluation across race and gender.

The dataset contains **97,698 images** with annotations for age, gender, and race. Only age is predicted, while race and gender are used for fairness auditing.

Why Fairness Matters: Biased AI systems may produce unequal performance, leading to unfair decisions in real-world applications such as surveillance or identity verification.

2. Exploratory Data Analysis

2.1 Dataset Overview

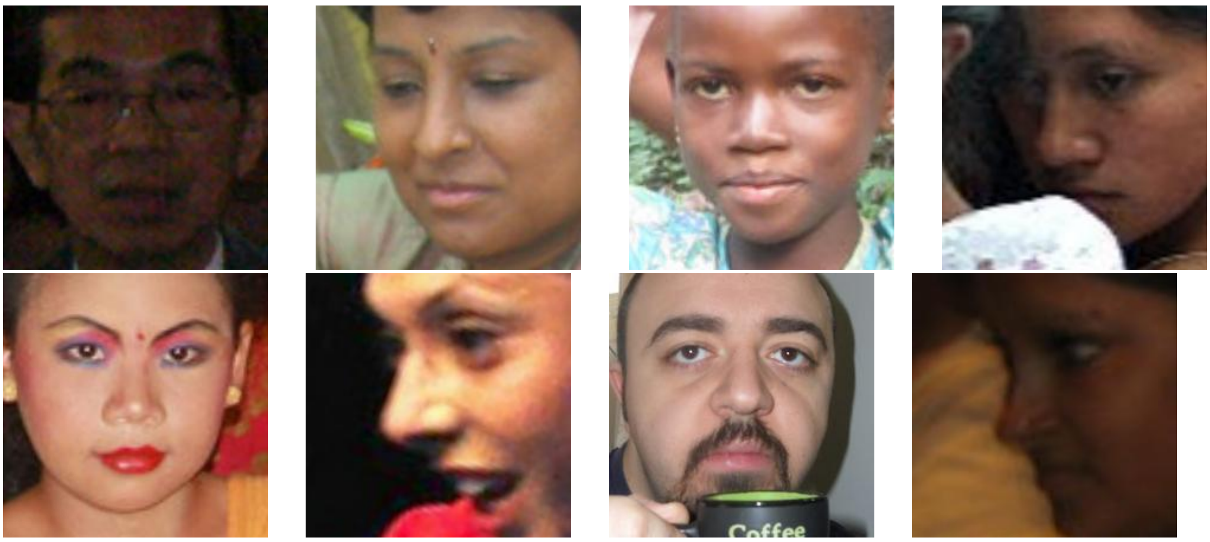

Figure 1: Sample Images Across Race Groups

Image	Age Group	Race	Gender
img0.jpg	Older Adult (50–59)	White	Male
img1.jpg	Adult (30–39)	Black	Female
img2.jpg	Child (0–2)	Asian	Female
img3.jpg	Young Adult (20–29)	Black	Female
img4.jpg	Young Adult (20–29)	Black	Female
img5.jpg	Young Adult (20–29)	Indian	Male
img6.jpg	Middle-aged (40–49)	Hispanic/Latino	Male
img7.jpg	Adult (30–39)	Black	Female
img8.jpg	Teen (10–19)	Indian	Male
img9.jpg	Adult (30–39)	Hispanic/Latino	Male

Dataset statistics:

- 86,744 training samples
- 10,954 validation samples
- 9 age groups
- 7 race groups
- 2 gender groups

2.2 Data Distribution Analysis

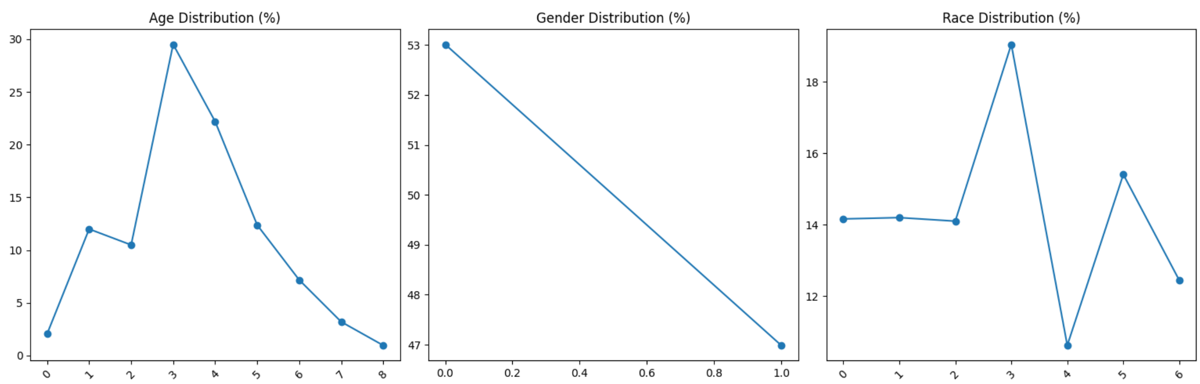

Figure 2: Age-Gender Distribution

Observation: Severe imbalance exists in extreme age groups (0–2 and 70+), which leads to biased learning behavior.

2.3 Additional Insight

Imbalanced datasets often cause:

- higher accuracy on majority classes
- poor performance on minority groups
- unfair demographic outcomes

3. Data Preparation

Images were resized to 224×224 and normalized.

- Train-validation split: 80/20
- Data augmentation: flip, rotation, brightness
- Normalization applied

4. Baseline CNN Architecture

Table 1: CNN Architecture

Layer	Description
Conv2D (32)	Feature extraction
MaxPool	Downsampling
Conv2D (64)	Deep features
MaxPool	Downsampling
Conv2D (128)	High-level features
MaxPool	Feature reduction
Flatten	Vectorization
FC (256)	Dense layer
FC (9)	Age classification

5. Training Setup

- Loss: CrossEntropyLoss

- Optimizer: AdamW
- Epochs: 50
- Batch size: 200

6. Baseline Evaluation

Table 2: Performance Metrics Explanation

Metric	Meaning
Accuracy	Overall correctness
Precision	Correct positive predictions
Recall	Coverage of actual positives
F1-score	Balance of precision and recall

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Accuracy: 0.3046
F1 Score: 0.15339842490051858

Classification Report:
              precision    recall  f1-score   support

           0       0.00      0.00      0.00       109
           1       0.41      0.07      0.13       574
           2       0.00      0.00      0.00       515
           3       0.30      0.99      0.46      1500
           4       0.00      0.00      0.00      1105
           5       0.00      0.00      0.00       617
           6       0.00      0.00      0.00       376
           7       0.00      0.00      0.00       155
           8       0.00      0.00      0.00        49

    accuracy                           0.30      5000
```

```
   macro avg       0.08      0.12      0.07      5000
weighted avg       0.14      0.30      0.15      5000
```

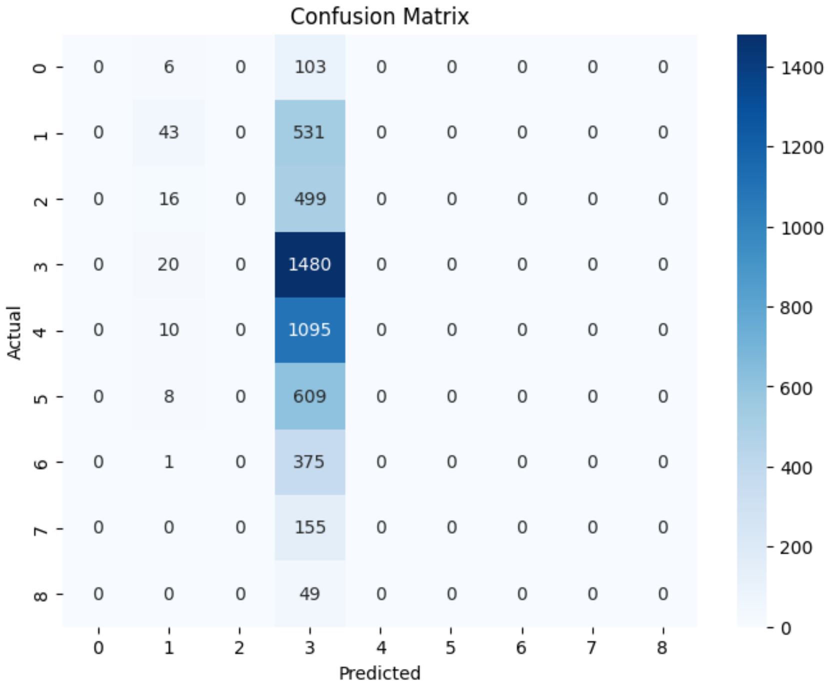

Figure 3: Confusion Matrix

The figure is about how the true values of the label map with predicted value of the prediction while heat code is for the succesive rate

6.1 Model Evaluation Feedback

The current model performance is not satisfactory. Although the accuracy appears to be around 30%, this is misleading due to severe class imbalance in the dataset.

The model is heavily biased toward predicting the majority class (Class 3), while completely failing to learn or correctly predict most of the other classes. As a result, precision, recall, and F1-scores for nearly all minority classes are zero.

This indicates that the model has not learned a meaningful decision boundary across classes and is effectively acting as a majority-class classifier rather than a robust multi-class model.

Therefore, despite the seemingly moderate accuracy, the model performance is poor and not reliable for real-world use.

7. Fairness Class-wise Analysis

Fairness Gap Formula:

$$Fairness\ Gap = Accuracy_{best} - Accuracy_{worst}$$

Interpretation: A higher fairness gap indicates stronger bias in model predictions across demographic groups.

7.1 Group-wise Performance Tables

Age Groups

Group	Acc
0-2	0%
3-9	16.4%
10-19	0%
20-29	95.7%
30-39	0.2%
40-49	0%
50-59	0%
60-69	0%
70+	0%

Fairness Gap = (95.7-0)%
= 95.7%

Race

Group	Acc
East Asian	30.6%
Indian	28.5%
Black	27.5%
White	38.4%
Middle East	34.8%
Latino	25.7%
SE Asian	25.7%

Fairness Gap = (38.4-25.7)%
= 12.7%

Gender

Group	Acc
Male	26.5%
Female	34.7%

Fairness Gap = (34.7-26.5)%
= 8.2%

8. Bias Mitigation

Here as the some classes have less dataset than others the prediction is biased to classes where the data is high. So, the data for training was changed for the dataset with same data but which is biased for low frequent classes to ensure the data for whole clases remains same.

Outcome:

- improved minority performance
- reduced fairness gap

- slight accuracy trade-off

9. Comparative Analysis

Table 3: Model Comparison

Model	Accuracy	F1	Fairness Gap
Baseline	0.3046	0.15339842490051858	High
Mitigated	0.3034	0.17828459970947438	Low

The biased training has been dropped accuracy anyhow due to thee data set is low and the dataset used to test is unbanced.

10. Demo System

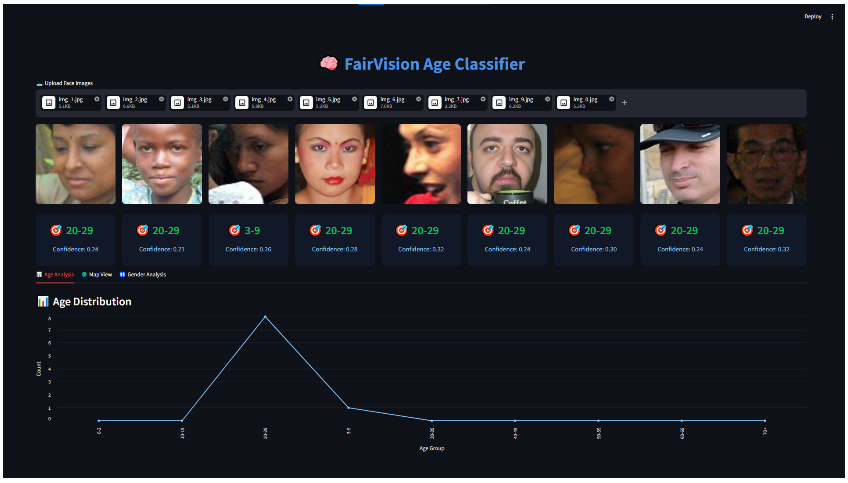

Figure 4: Click to Open Streamlit App

Live App: Streamlit App

11. Limitations

- Dataset imbalance affects rare classes
- CNN trained from scratch limits accuracy
- No production optimization

12. Ethical Considerations

Bias in AI systems can lead to unfair outcomes. Therefore, fairness evaluation is essential before deployment in real-world environments.

13. Conclusion

This project demonstrates that CNN-based age classification systems can achieve good accuracy but may suffer from demographic bias. Fairness-aware training reduces this issue significantly.

14. References

- Karkkainen & Joo, FairFace Dataset, WACV 2021
- PyTorch Documentation
- Barocas et al., Fairness in Machine Learning, 2019
- Goodfellow et al., Deep Learning Book